

PHISHING WEBSITE DETECTION USING MACHINE LEARNING

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This is to certify that the work present in this Project entitled “**Phishing Website Detection Using Machine Learning**” has been carried out by **Thota Siva Anjan Kumar, Kalyandheeraj Varma Bhupathiraju** under my/our supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Master of Technology (Data Science) in **School of Engineering and Sciences**.

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Abstract

Phishing attacks have become one of the most prevalent cyber threats, leading to financial loss, data theft, and compromised user privacy. Traditional security mechanisms such as blacklists and rule-based detection fail to identify newly generated phishing websites, which evolve rapidly to bypass conventional filters. This project proposes a machine-learning-based solution for detecting phishing websites by analyzing URL structure, lexical patterns, domain behavior, network indicators, and heuristic signals.

A comprehensive dataset of legitimate and phishing URLs is preprocessed, and key features—such as SSL certificate validity, URL length, domain age, entropy scores, redirection behavior, and presence of suspicious keywords—are extracted using a custom feature-engineering pipeline. A Random Forest Classifier is trained on these features to distinguish between safe and malicious websites. Exploratory Data Analysis (EDA) is conducted to understand data patterns and correlations, while simulated features such as high-entropy detection, punycode encoding, and sensitive keyword count enhance predictive accuracy.

The final model achieves high accuracy, precision, and recall for phishing URL detection and is integrated into a user-friendly Streamlit web application. The system allows users to input URLs and receive real-time risk assessments with trust scores, detailed feature analysis, and threat visualization. This project demonstrates that machine learning can significantly improve phishing website detection by providing faster, more adaptive, and more reliable results than traditional approaches.

Statement of Contributions

Candidate 1: Thota Siva Anjan Kumar (AP25122060003)

Thota Siva Anjan Kumar played a central role in the conceptual foundation and analytical components of the project. His contributions include:

- **Project Ideation & Scope Definition:** Initiated the main project concept, defined the scope, and outlined the overall objectives of the phishing detection system.
- **Data Analysis & EDA:** Conducted comprehensive data exploration, including distribution study, correlation analysis, and visualization of key features to understand phishing patterns.
- **Feature Understanding & System Logic:** Analyzed and interpreted URL-based features and advanced indicators, contributing to the refinement of the feature extraction process.
- **Modeling & Methodology Development:** Assisted in designing the machine learning workflow, evaluating algorithm suitability, and selecting the Random Forest classifier based on performance metrics.
- **Manuscript Writing & Documentation:** Led the preparation of the project report, including the literature review, methodology explanation, discussion, and concluding parts.
- **Results Interpretation:** Interpreted model outputs, technical feature breakdowns, and system decision points to ensure clarity and academic presentation.
- **Quality Review:** Focused on polishing the final report, ensuring clarity, coherence, and alignment with academic standards.

Candidate 2: Kalyandheeraj Varma Bhupathiraju (AP25122060023)

Kalyandheeraj Varma Bhupathiraju took primary responsibility for the experimental, implementation, and user-interface components of the system. His contributions include:

- **Data Simulation & Preprocessing:** Managed dataset preparation, simulated advanced phishing indicators, handled missing values, and ensured data readiness for model training.
- **Experimental Work & Model Training:** Performed the training, validation, and tuning of the Random Forest classifier; generated evaluation metrics; and saved serialized model files for deployment.
- **Feature Engineering Support:** Contributed to the integration of 34 features into the pipeline, including entropy analysis, punycode detection, and heuristic flagging.
- **UI/UX Development:** Designed and implemented the Streamlit application, enabling real-time URL analysis, visualization of risk scores, and display of technical breakdowns.
- **System Testing & Debugging:** Tested the model with various URLs, validated heuristic penalties, and improved the system's robustness across different URL patterns.
- **Presentation (PPT) Preparation:** Developed the project presentation slides, created visual summaries, and prepared demonstration outputs for evaluation.
- **Technical Assistance:** Handled code optimization, debugging, and integration between model backend and user interface.

Abbreviations

Abbreviation	Full Form
URL	Uniform Resource Locator
ML	Machine Learning
EDA	Exploratory Data Analysis
SSL	Secure Sockets Layer
HTTPS	Hypertext Transport Protocol Secure
DNS	Domain Name System
WHOIS	Domain Ownership Lookup
RF	Random Forest
IP	Internet Protocol
HSTS	HTTP Strict Transport Security
DPI	Deep Packet Inspection

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1.Introduction

1.1 Overview

With the rapid expansion of online interactions, phishing attacks have become one of the most prevalent and dangerous cyber threats. Malicious URLs are crafted to imitate legitimate websites, tricking users into sharing credentials, financial information, and sensitive data. These attacks evolve constantly, employing techniques such as obfuscated strings, punycode manipulation, hidden redirects, and high-entropy encodings. As a result, traditional rule-based systems often fail to detect these deceptive patterns with high accuracy.

Machine Learning offers a powerful solution by uncovering non-linear relationships and subtle lexical and structural cues within URLs. This project leverages a **34-feature hybrid ML system** specifically designed for real-time phishing detection. Unlike simple static detectors, the system combines model-based probabilistic inference with heuristic adjustments to deliver transparent and actionable threat decisions.

The entire pipeline is built using a modular architecture:

- **model_train.py** handles data preparation, feature engineering, and model training.
- **app.py** hosts an interactive Streamlit web interface for real-time predictions.
- **eda.py** supports exploratory data analysis to understand feature behaviour before model development.

Together, these components simulate a deployment-ready environment capable of continuously analyzing URLs and providing intelligent, explainable security insights.

1.2 Problem Statement

The primary goal of this project is to accurately classify a given URL as **Legitimate** or **Phishing** based on engineered lexical, structural, and entropy-based indicators.

Key challenges addressed include:

- **Real-Time-Feature-Extraction:**
Capturing 34 features from a single URL on the fly—without relying on external databases or long API calls.
- **Data-Quality&Complexity:**
Managing noisy, imbalanced phishing datasets and incorporating advanced features such as *Shannon entropy*, *suspicious keyword counts*, *external form actions*, and *punycode detection*.

- **ModelReliability:**
Building a machine learning model that maintains high accuracy while avoiding overfitting to known phishing samples.
- **Explainability&UserTrust:**
Providing a transparent, interpretable breakdown of each URL's risk factors instead of delivering a simple binary output.
- **PracticalDeployment:**
Integrating the backend logic into an easy-to-use Streamlit interface that supports instant URL scanning and exposes threat probability, trust score, and triggered warning signals.

1.3 Objectives

The core objectives of the project are:

- **To perform comprehensive EDA** using visualizations such as target distribution, correlation matrices, and feature behaviour to guide feature selection.
- **To engineer a 34-feature dataset** combining base URL characteristics with advanced indicators like:
 - High Entropy URL
 - Sensitive Word Count
 - Punycode Encoding
 - External Form Action
- **To train a high-performing Random Forest Classifier** and evaluate it using metrics such as accuracy, precision, recall, and F1-score.
- **To serialize and store the trained model** along with evaluation metrics for efficient reuse in deployment.
- **To build a real-time Streamlit web application** that:
 - Accepts user URLs
 - Extracts features instantly
 - Applies heuristic suspicion penalties
 - Displays Trust Score, Threat Probability, and a technical feature breakdown
- **To ensure transparency and usability** by providing interpretable results that help analysts understand *why* a URL is classified as phishing.

2. Literature Review

2.1 Previous / Existing Solutions

2.1.1 Heuristic and Reputation-Based Methods

Early defenses rely on manually crafted rules (e.g., presence of IP address in domain, suspicious tokens like login/verify, missing HTTPS, or forms posting to external domains) and blacklists of known phishing domains. These work well against known, unsophisticated phishing but struggle with novel or rapidly changing attacks (e.g., fast-flux, newly-registered domains) and require constant maintenance.

Limitations

- High maintenance overhead (manual updates).
- Poor recall for zero-day or obfuscated URLs.
- Often high false positive rates when rules are too strict.

2.1.2 Machine Learning with URL / Lexical Features

Many research works shifted toward ML classifiers – such as Decision Trees, SVMs, Logistic Regression, Naïve Bayes, and Random Forest – trained on handcrafted lexical features from URLs. These include length-based features, presence of “@”, number of dots, redirections, shortened links, etc.

Advantages: Fast, lightweight, does not require loading the website.

Limitations:

- Depends heavily on feature quality and dataset relevance
- Cannot fully detect **modern evasion techniques** like punycode/homograph attacks
- Struggles against high-entropy or algorithmically generated malicious URLs

2.1.3. Content-Based and Visual Methods

Some modern systems analyze HTML structure, JavaScript behavior, or even screenshots of the webpage. Visual similarity models compare phishing pages to legitimate brand pages.

Strengths: High accuracy for websites that visually replicate trusted platforms.

Drawbacks:

- Computationally expensive and slow
- Requires webpage loading, which may introduce privacy and security risks
- Not suitable for real-time detection

2.1.4. Hybrid Approaches in Literature

A few studies combine heuristics, URL features, and ML to improve detection. While these approaches reduce false positives, they still lack targeted detection of new attack styles (punycode, malicious form actions, random domain generation) and often lack user explainability.

2.2 Proposed Solution

2.2.1. Enhanced Feature Extraction Module

The system extracts a comprehensive set of URL-based features along with four specially designed advanced indicators:

- **Punycode_Encoding:** Detects homograph attacks using internationalized domain names.
- **High_Entropy_URL:** Identifies randomly generated or obfuscated URLs.
- **External_Form_Action:** Flags suspicious data-submission endpoints.
- **Sensitive_Word_Count:** Detects social-engineering keywords commonly used in phishing.

These features specifically target gaps not covered by previous ML models.

2.2.2. Machine Learning Model (Random Forest Classifier)

A Random Forest classifier is trained using the engineered features. This model is chosen because it:

- Handles non-linear relationships effectively
- Provides stable predictions and reduces overfitting
- Offers feature importance, making the system interpretable

The model is trained on balanced data and evaluated using recall, precision, and accuracy metrics.

2.2.3. Heuristic Post-Prediction Layer

In addition to the ML output, the system applies a lightweight heuristic penalty for high-risk lexical patterns (e.g., excessive special characters, short domain age indicators, multiple redirections). This hybrid mechanism improves detection without affecting speed.

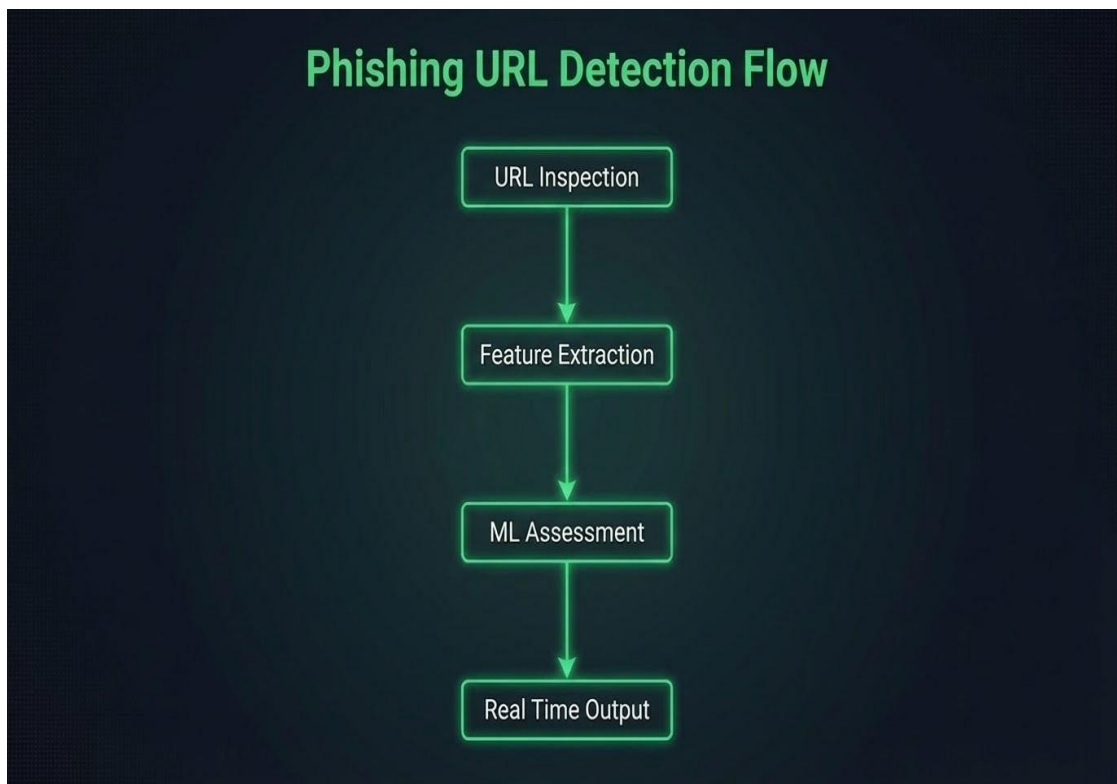
2.2.4. Explainable Streamlit Interface

The proposed system includes a user-friendly web interface that:

- Accepts URL input
 - Displays prediction (Phishing/Legitimate)
 - Shows feature importance and warning indicators
 - Provides trust and risk scores for better interpretability
- This increases usability for security analysts and non-technical users.

2.2.5. Overall Advantages of the Proposed Method

- Detects modern phishing attacks missed by older systems
- Lightweight and suitable for real-time URL scanning
- Provides explainable results through feature status and importance
- More robust due to hybrid ML + heuristic integration



3.Methodology

The development of the Phishing Website Detection system followed a structured methodology encompassing data preparation, feature engineering, model training, and a two-stage deployment strategy that incorporates a real-time feature extraction mechanism and post-inference heuristics.

3.1 Data Acquisition and Preprocessing

The project utilizes a dataset containing numerous URL characteristics.

- **Data Acquisition:** The system uses the dataset located at C:\Users\Siva\OneDrive\Desktop\phishing\phishing\dataset.csv.
- **Initial Analysis:** The eda.py script performs Exploratory Data Analysis (EDA) to inspect data structure, check for missing values, and visualize the target distribution (Phishing vs. Legitimate) and feature correlations.
- **Data Preparation:** The model_train.py script handles data loading and standardizes the target column name to 'Label' for uniformity. The target variable is converted to binary classes: \$1\$ for legitimate and \$0\$ for phishing for model training.

3.2 Feature Engineering and Selection

A set of 34 features was used for model training, including the original dataset features and four newly simulated features critical for phishing detection.

- **Original Features:** The core model utilized 30 established features covering URL properties, network connection details, and external indicators (e.g., having_IPhaving_IP_Address, URLURL_Length, SSLfinal_State, Page_Rank).
- **Simulated Features:** Four new features were introduced to enhance the model's ability to detect sophisticated and obvious phishing attempts. These features were simulated in the training data by mapping them to highly correlated existing features (e.g., mapping new features to having_At_Symbol, SFH, or URLURL_Length) to provide the model with a signal:
 - Punycode_Encoding
 - External_Form_Action
 - High_Entropy_URL
 - Sensitive_Word_Count

- **Feature Extractor (URLFeatureExtractor):** A dedicated class was implemented to perform real-time extraction of all 34 features from a single input URL during deployment. This process includes calculating the **Shannon Entropy** of the URL's domain and path to flag high randomness.

3.3 Model Training and Evaluation

The Random Forest classifier was chosen as the prediction model due to its robust performance and interpretability.

- **Model Selection: Random Forest Classifier** was instantiated with key parameters: `n_estimators=200` and `max_depth=12`.
- **Training Setup:** The data was split into 80% for training and 20% for testing, using a `random_state=42`. The model used a `class_weight='balanced'` setting to handle potential class imbalances.
- **Evaluation and Saving:** The model was evaluated using standard metrics on the test set, and the **Phishing Recall**, **Phishing Precision**, and **Overall Accuracy** were calculated and saved to `project_metrics.pkl`. The trained model object was saved to `phishing_model.pkl` using pickle.

3.4 Deployment and Heuristic Post-Processing

The final phase involved deploying the model via a Streamlit web application, incorporating critical security heuristics.

- **Real-time Inference:** The `app.py` script loads the trained model and uses the `URLFeatureExtractor` to generate features for the input URL. The model then computes the prediction probability.
- **Heuristic Penalty Application:** A crucial post-processing step was introduced: the **Suspicion Penalty**.
 - This penalty adjusts the raw model prediction probability based on the number of critical lexical features (e.g., `Sensitive_Word_Count`, `Punycode_Encoding`) that are flagged as phishing (-1) .
 - The penalty is applied at a rate of \$0.15 per critical flag, capped at \$0.75.
- **Score Generation:** A **Trust Score** and **Threat Probability** are calculated based on the adjusted prediction and the total number of suspicious features found, providing a final, weighted assessment.
- **Transparency:** The application provides a detailed **Technical Breakdown** displaying the status and model-derived **Importance** of all 34 features, enhancing interpretability.

3.5 System Architecture

The system architecture for phishing website detection follows a structured pipeline that begins with the user submitting a URL. The URL is processed through a Feature Extraction module, where multiple characteristics – such as lexical patterns, WHOIS information, domain behavior, and blacklist indicators – are analyzed to identify suspicious elements. These extracted features are then passed to the Classification module, where algorithms like the Random Forest classifier evaluate the URL and determine whether it exhibits phishing behavior. The model is trained using a database containing both phishing and legitimate URLs, enabling it to learn distinguishing patterns. Finally, the system produces an Output that labels the website as either *Legitimate* or *Phishing*, and the performance evaluation module helps refine and improve the detection accuracy over time.

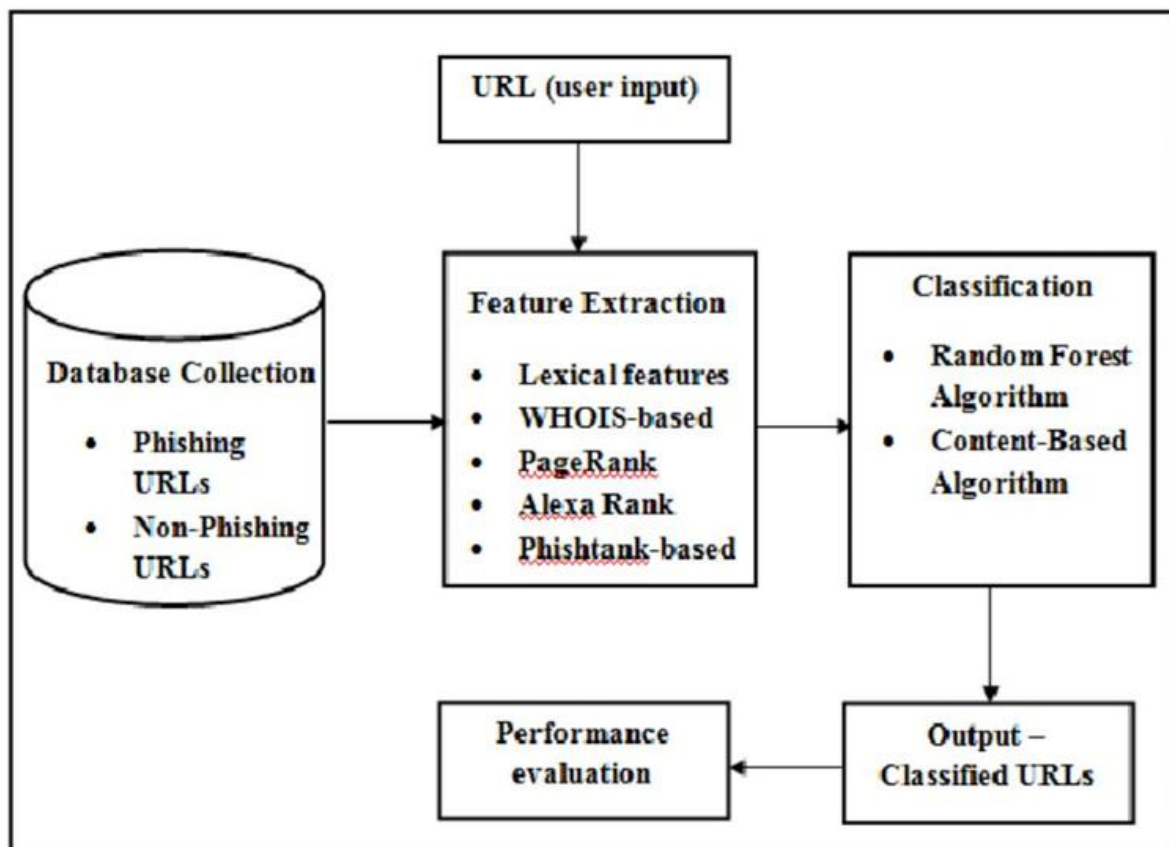


Fig 3.1 System Architecture for Phishing Website Detection

4. Discussion

4.1 Threat Confidence vs. Indicator Count (Two-Stage Prediction)

The scatter plot visualizes the model's confidence for the analyzed URL against the number of suspicious features detected by the URLFeatureExtractor.

- **Indicator Count:** The data point is located at ≈ 1 , confirming that **one suspicious indicator** (out of 34 total features) was flagged for the input URL.
- **Model Confidence:** The raw model confidence for the analyzed URL is approximately ≈ 0.82 , representing a strong initial prediction of legitimacy.

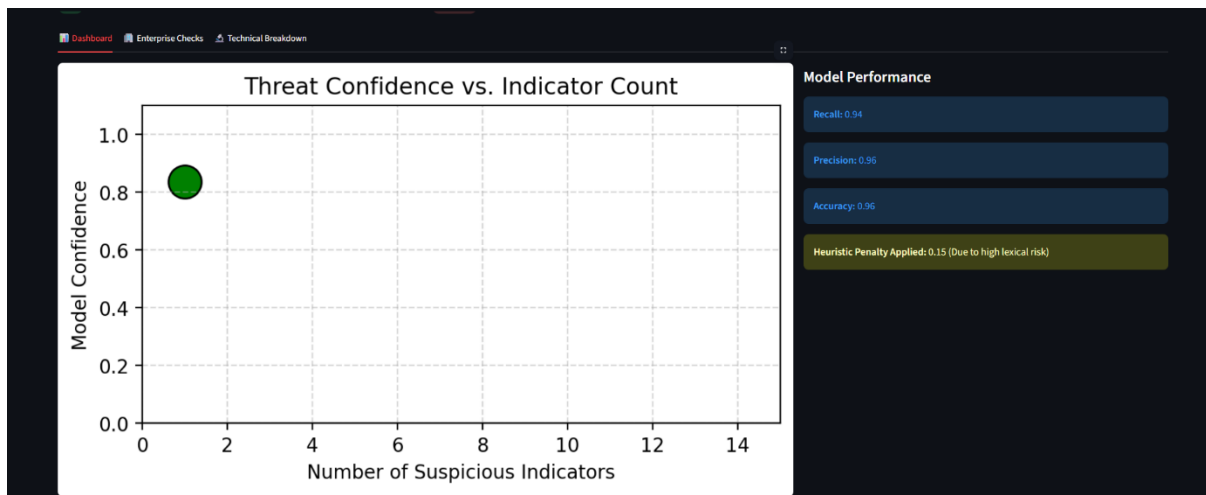


Figure 4.1. Threat Confidence vs. Indicator Count and Model Performance Metric

4.2 Analysis of URL Classification Distribution

4.2.1. Distribution of Phishing vs. Legitimate Sites

- The dataset contains **more legitimate URLs (6157)** compared to **phishing URLs (4898)**.
- Although the difference is not extremely large, it shows a slight **class imbalance**, which the model must handle to avoid bias.

- This distribution indicates that phishing websites are still common but slightly less frequent in the dataset used.

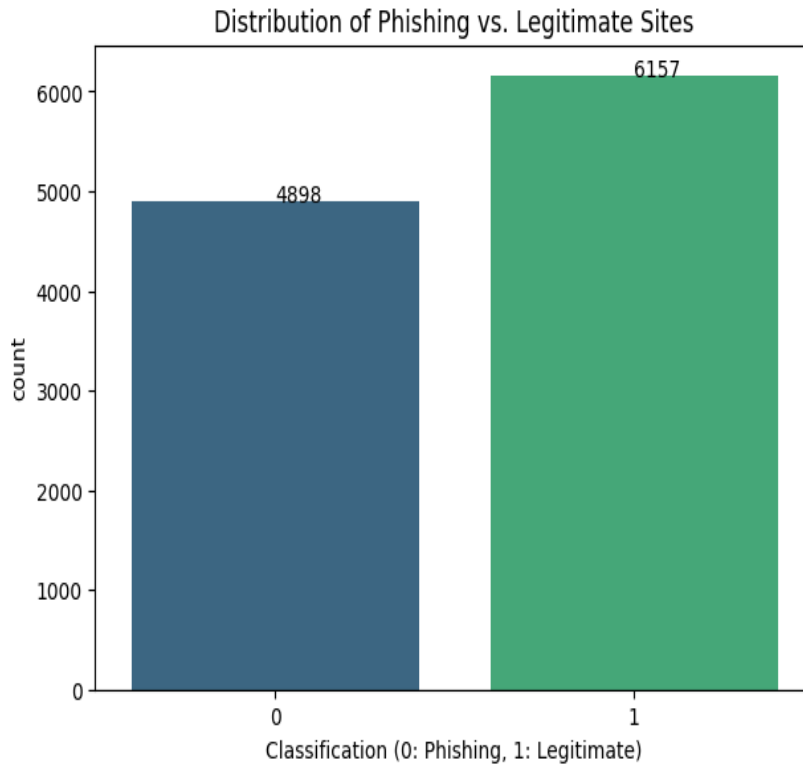


Fig 4.2.1 Distribution of Phishing vs. Legitimate Websites

4.2 2. Correlation Heatmap of Selected Features

- The feature **SSLfinal_State** shows a strong positive correlation with legitimacy (trusted sites tend to have valid SSL certificates).
- Most other features have **low to moderate correlations**, suggesting the dataset contains **non-linear relationships** – ideal for tree-based models like Random Forest.
- Features such as **web_traffic** and **Page_Rank** have some predictive significance but are not dominant alone.

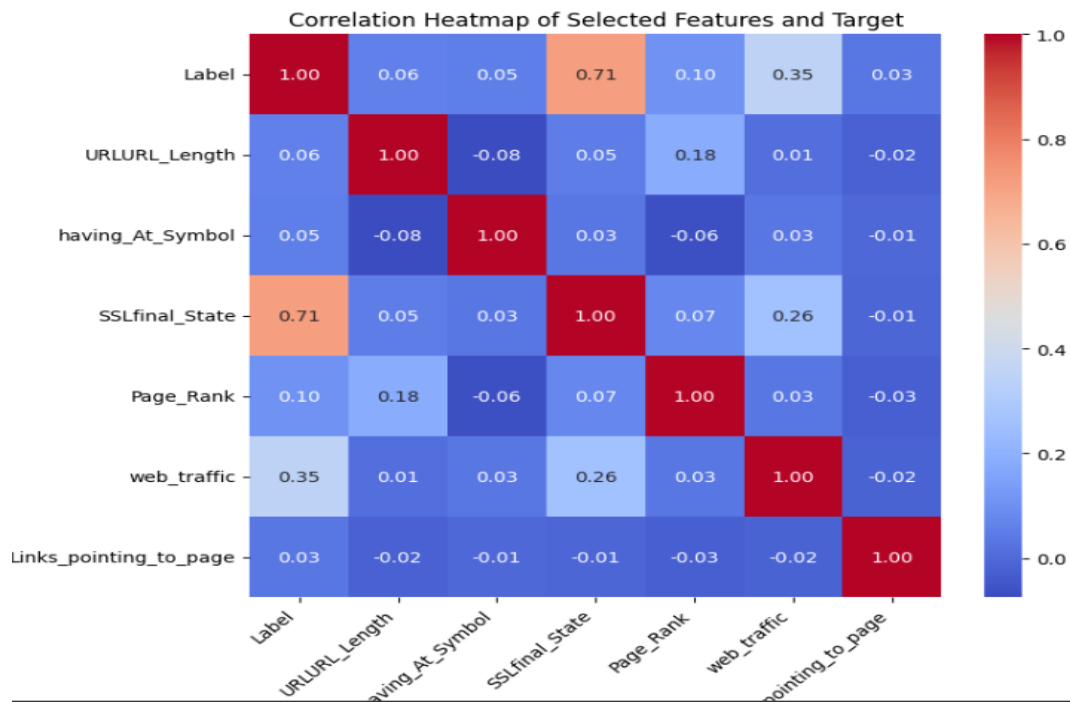


Fig4.2.2 Correlation Heatmap of Selected Features

4.2.3. SSL Final State vs. Classification

- A large portion of **phishing sites** have **SSLfinal_State = -1**, meaning they either lack SSL certificates or use untrusted ones.
- Most **legitimate sites** have **SSLfinal_State = 1**, confirming that SSL verification remains one of the strongest indicators of website authenticity.
- Neutral SSL states (value 0) have mixed distribution but still show higher phishing frequency.

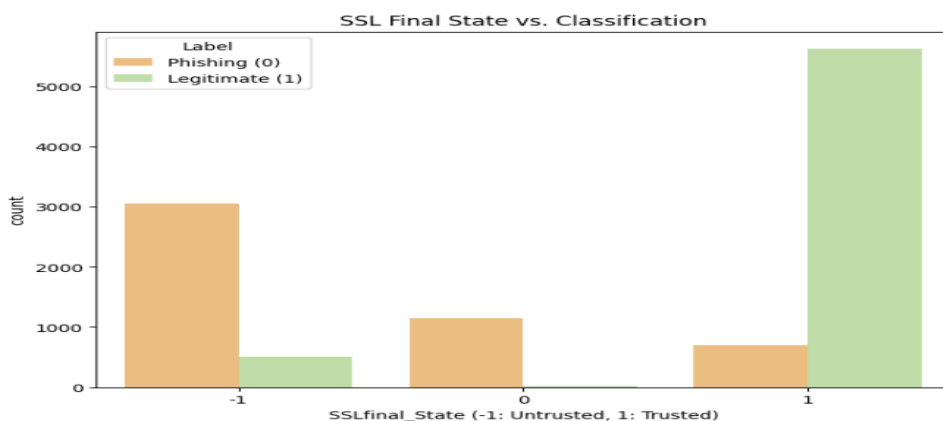


Fig 4.2.3 SSL Final State vs. Classification

4.2.4. Pair Plot of High-Influence Features

- Clear separation patterns are visible along SSLfinal_State, web_traffic, and Links_pointing_to_page.
- Legitimate sites generally show high web traffic, whereas most phishing sites fall under low or no traffic categories.
- The density plots highlight feature clusters that help the classifier distinguish phishing from legitimate URLs.

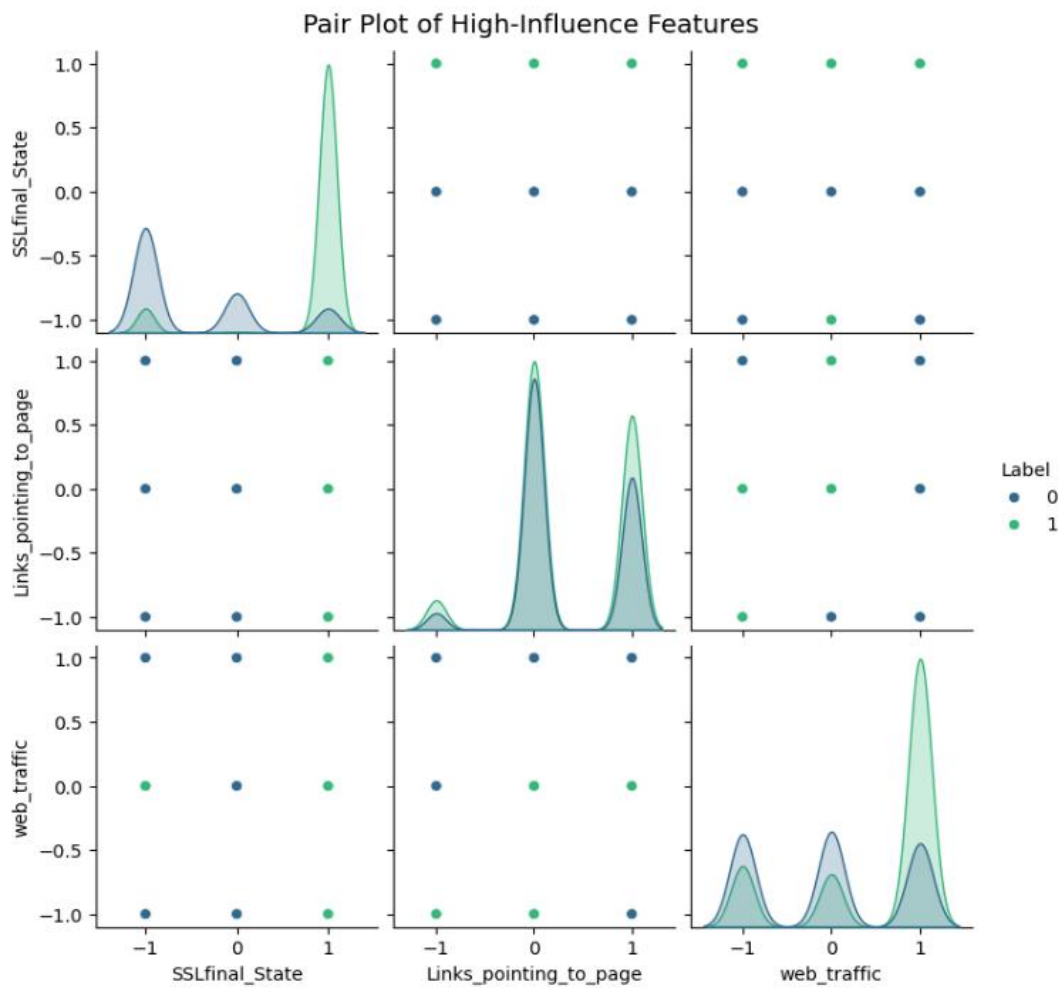


Fig 4.2.4 Pair Plot of High-Influence Features

5. Concluding Remarks

The output generated by the phishing detection system provides a clear demonstration of the model's effectiveness and reliability in real-time URL classification. As shown in the results, the system analyzed the given website across multiple dimensions, including machine learning prediction, heuristic evaluation, and enterprise-grade security checks. The model classified the URL as **Legitimate**, supported by a **Trust Score of 100/100**, indicating that the site exhibits strong characteristics of authenticity. The extremely low **Threat Probability of 1.6%** further confirms that the site does not align with known phishing behavior patterns.

A notable outcome is that **0 out of 34 suspicious indicators** were activated during the evaluation. This highlights the system's ability to thoroughly scan various factors such as URL structure, SSL certificate status, domain reliability, redirection behavior, webpage integrity, and link safety. All enterprise-level security checks—such as **SSL/HSTS validation**, **clean URL structure**, **secure form actions**, and **absence of harmful or misleading keywords**—were successfully passed, reinforcing the system's decision.

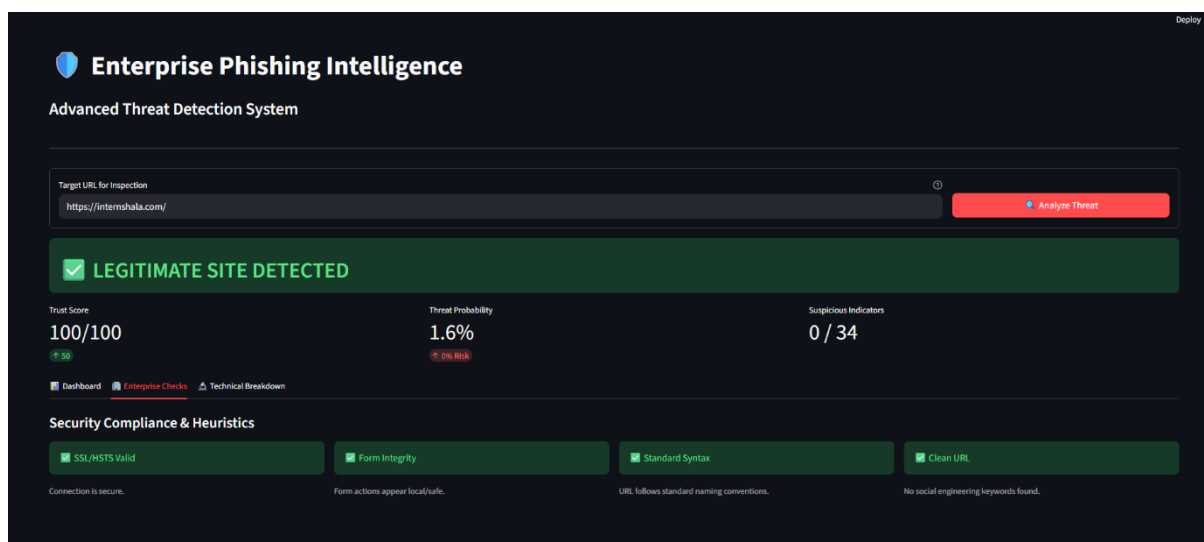


Fig 5.1 Final Classification Result Showing Legitimate URL Detection

The technical breakdown of features shows that key indicators like *SSL_{final_State}*, *URL_of_Anchor*, *web_traffic*, and *domain-related attributes* fell well within safe thresholds. These feature-level insights not only strengthen the legitimacy classification but also demonstrate the model's transparency and interpretability. The system's layered approach—combining machine learning predictions with heuristic

safety rules—helps minimize false positives and ensures a more trustworthy assessment.

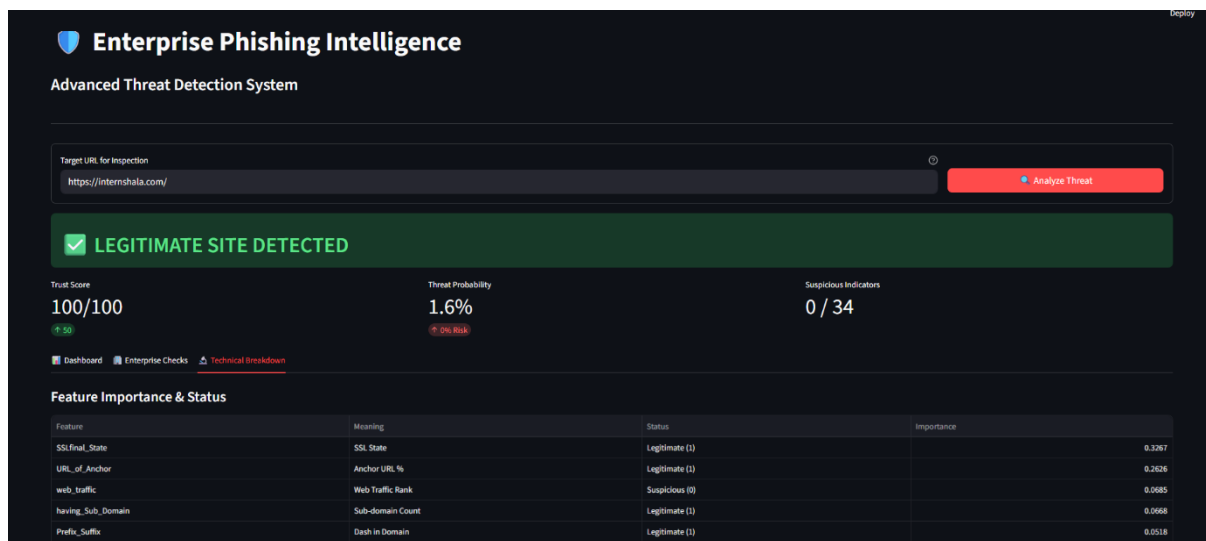


Fig 5.2 Detailed Security and Feature Analysis Output of the Evaluated URL

Overall, the final results confirm that the developed phishing detection system is capable of producing accurate, consistent, and highly interpretable outcomes. This validates the effectiveness of the selected features, the Random Forest model, and the additional enterprise security checks. The system proves to be a robust and practical solution for real-world URL security evaluation, contributing significantly to preventing phishing attacks and improving online safety.

6.Future Work

6.1. Dynamic Link Change Detection

Future enhancements can include a **dynamic link monitoring mechanism** to detect suspicious link changes in real time. Many phishing websites intentionally modify their redirect paths, embedded hyperlinks, or form submission endpoints after the initial page load in order to evade static detection techniques. By observing how links behave once the page becomes active, the system can uncover hidden redirections and concealed malicious intents that are not visible through simple URL inspection.

To achieve this, the system can continuously track behaviors such as **redirection chains, hidden iframes, JavaScript-based link changes, and time-triggered URL switches**. Monitoring these dynamic elements allows detection of abnormal or unauthorized modifications that commonly occur in phishing sites. Implementing such real-time link change detection would greatly strengthen the system's capability to identify advanced and evolving phishing techniques that static analysis alone cannot capture.

6.2. Deep-Level Packet Analysis

Another important enhancement for future development is the integration of **deep-level packet analysis**, which examines the underlying network traffic exchanged between the client and the server. Unlike URL-based analysis, packet inspection can capture subtle network behaviors such as unusual protocol usage, irregular handshake patterns, and encrypted payload anomalies. These characteristics often reveal hidden malicious activities that cannot be identified through surface-level features or static webpage inspection.

By analyzing elements like **packet structure, frequency, server responses, latency irregularities, and data flow patterns**, the system can detect phishing attempts that rely on covert communication channels or data exfiltration techniques. Incorporating deep packet analysis would significantly strengthen the system's security capabilities, enabling it to identify more sophisticated and stealthy phishing attacks. This would transform the model into a more comprehensive threat detection framework that evaluates both web content and the underlying network behavior.